

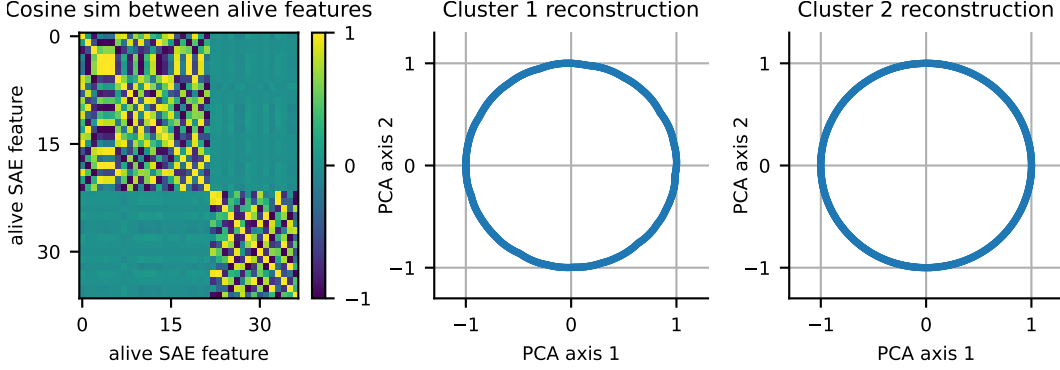


Figure 13: Automatic discovery of synthetic circular features by clustering SAE dictionary elements.

on layers 8, 16, and 24 out of 32 total layers to maximize coverage of the model’s representations. We use a $16\times$ expansion factor, yielding a total of 65536 dictionary elements for each SAE.

To train our SAEs, we use an L_p sparsity penalty for $p = 1/2$ with sparsity coefficient $\lambda = 0.012$. Before an SAE forward pass, we normalize our activation vectors to have norm $\sqrt{d_{model}} = 64$ in the case of Mistral. We do not apply a pre-encoder bias. We use an AdamW optimizer with weight decay 10^{-3} and learning rate 0.0002 with a linear warm up. We apply dead feature resampling (Bricken et al., 2023) five times over the course of training to converge on SAEs with around 1000 dead features.

F GPT-2 AND MISTRAL 7B DICTIONARY ELEMENT CLUSTERING

In this section, we first present pseudocode in Alg. 1 for the overall high level technique that finds multi-dimensional features and that uses clustering as a subroutine. We then provide the specific clustering algorithm implementations we use for GPT-2 and Mistral.

Algorithm 1: High Level Clustering Approach For Finding Multi-D Features

Input: Dictionary elements D , activation vectors $X_{i,l}$, SAE

Output: Irreducible multi-dimensional features

```

 $S_{i,j} \leftarrow \text{CosineSim}(D_i, D_j);$ 
 $clusters \leftarrow \text{Cluster}(S);$ 
 $reconstructions \leftarrow \{\};$ 
for  $cluster$  in  $clusters$  do
     $R_{cluster} \leftarrow \text{ids of dictionary elements in } cluster;$ 
    for  $x_{i,l}$  in  $X_{i,l}$  do
         $encoding \leftarrow \text{ReLU}(E \cdot x_{i,l});$ 
        if  $\max(encoding[R_{cluster}]) > 0$  then
             $r \leftarrow D[:, R_{cluster}] \cdot encoding;$ 
             $reconstructions \leftarrow reconstructions \cup \{r\};$ 
        end
    end
end
 $features \leftarrow \{\};$ 
for  $R$  in  $reconstructions$  do
     $proj \leftarrow \text{PCA}(R);$ 
    if  $\text{TestIrreducible}(proj)$  then
        Add  $proj$  to  $features$ ;
    end
end
return  $features$ 

```

F.1 GPT-2-SMALL METHODS AND RESULTS

For GPT-2-small, we perform spectral clustering on the roughly 25k layer 7 SAE features from (Bloom, 2024), using pairwise angular similarities between dictionary elements as the similarity matrix. We use `n_clusters = 1000` and manually looked at roughly 500 of these clusters. For each cluster, we looked at projections onto principal components 1-4 of the reconstructed activations for these clusters. In Fig. 14, we show projections for the most interesting clusters we identified, which appear to be circular representations of days of the week, months of the year, and years of the 20th century.

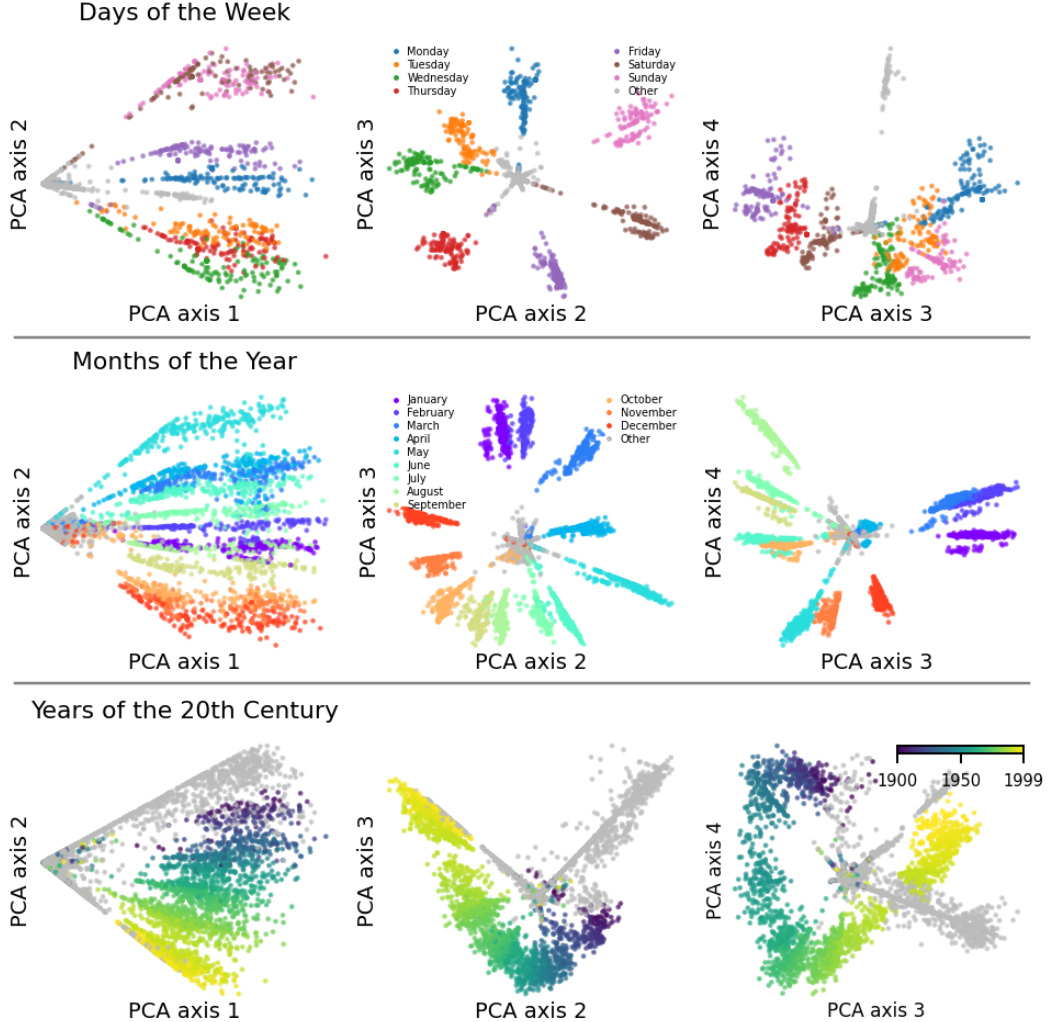


Figure 14: Projections of days of week, months of year, and years of the 20th century representations onto top four principal components, showing additional dimensions of the representations than Fig. 1.

F.2 MISTRAL 7B METHODS AND RESULTS

For Mistral 7B, our SAEs have 65536 dictionary elements and we found it difficult to run spectral clustering on all of these at once. We therefore develop a simple graph based clustering algorithm that we run on Mistral 7B SAEs:

1. Create a graph G out of the dictionary elements by adding directed edges from each dictionary element to its k closest dictionary elements by cosine similarity. We use $k = 2$.

2. Make the graph undirected by turning every directed edge into an undirected edge.
3. Prune edges with cosine similarity less than a threshold value τ . We use $\tau = 0.5$.
4. Return the connected components as clusters.

We run this algorithm on the Mistral 7B layer 8 SAE (2^{16} dictionary elements) and find roughly 2700 clusters containing between 2 and 1000 elements. We manually inspected roughly 2000 of these. From these, we re-discover circular representations of days of the week and months of the year, shown in Fig. 15. However, we did not find other obviously interesting and clearly irreducible features.

We also investigate the sensitivity of this method to τ and k by varying τ and k and showing the max Jaccard similarity between any of the resulting clusters and the days of the week cluster we show in Fig. 15. We show the results in Fig. 16, where we find that varying k has minimal effect, while varying τ shows 3 regimes: small τ causes all features to group in one cluster, so the days of the week cluster is not found; medium τ causes the days of the week cluster to become identifiable; large τ causes all features to be divided into their own clusters.

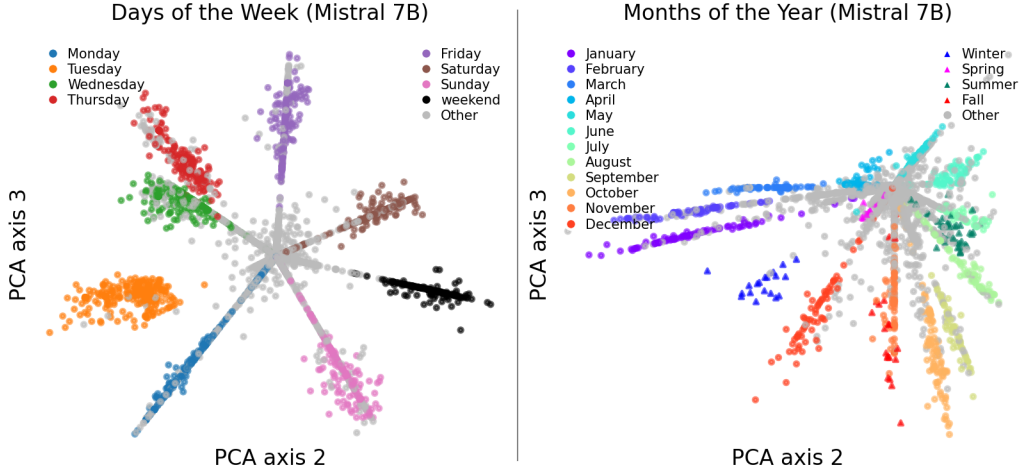


Figure 15: Circular representations of days of the week and months of the year which we discover with our unsupervised SAE clustering method in Mistral 7B. Unlike similar features in GPT-2, we also find an additional “weekend” representation in between Saturday and Sunday representations (left) and additional representations of seasons among the months (right). For instance, “winter” tokens activate a region of the circle in between the representation of January and December.

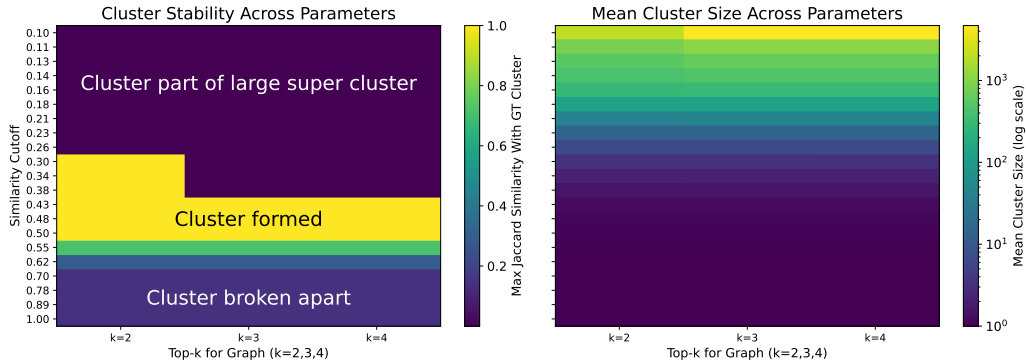


Figure 16: Hyperparameter regimes where the days of the week cluster exists. The cluster exists in the regime between all features clumping together and all features being in their own cluster; this regime seems reasonably stable.

As future work, we think it would be exciting to develop better clustering techniques for SAE features. Our graph based clustering technique could likely be improved by more recent efficient and high-quality graph based clustering techniques, e.g. hierarchical agglomerate clustering with single-linkage (Lattanzi et al., 2020). Additionally, we believe we would see a large improvement by setting edge weights to be a combination of both the cosine and Jaccard similarity of the dictionary elements, e.g. $\max(\text{cosine}, \text{Jaccard})$.

G OTHER DISCOVERED CLUSTERS

In Fig. 17, we plot the top 11 ranked clusters by the product of a) the measured separability index and b) one minus the measured ϵ -mixture index with $\epsilon = 0.1$ (this is just one of many possible ways to get an ordered ranking from a two-parameter score). We color by both the current token (which results in clear patterns for all tokens) and the next token (to see if we find belief states as found by Shai et al. (2024) in toy transformers). We note that weekdays are ranked 9 and so are shown in the plot. Additionally, the next token patterns of the ‘such’ cluster and the ‘B’ cluster do seem to display some clustering independently of the the current token pattern, which might lend the belief state hypothesis some support.

H FURTHER EXPERIMENT DETAILS

H.1 ASSETS INFORMATION

We use the following open source models for our experiments: Llama 3 8B (AI@Meta, 2024) (custom Llama 3 license <https://llama.meta.com/llama3/license/>), Mistral 7B (Jiang et al., 2023) (released under the Apache 2 License), and GPT-2 (Radford et al., 2019) (modified MIT license, see <https://github.com/openai/gpt-2/blob/master/LICENSE>).

H.2 MACHINE INFORMATION

Intervention experiments were run on two V100 GPUs using less than 64 GB of CPU RAM; all experiments can be reproduced from our open source repository in less than a day with this configuration. We use the TransformerLens library (Nanda & Bloom, 2022) for intervention experiments. ϵ -mixture index measurements on toy datasets took about one minute each, on 8GB of CPU RAM. EVR experiments take seconds on 8GB of CPU RAM and are dominated by time taken to human-interpret the RGB plots.

GPT-2 SAE clustering and plotting was run on a cluster of heterogeneous hardware. Spectral clustering and computing reconstructions + plotting was done on CPUs only. We made reconstruction plots for 500 clusters, with each taking less than 10 minutes. Mistral 7B SAE reconstruction plots were made on the same cluster. We made roughly 2000 reconstruction plots for Mistral 7B (and manually inspected each), with each taking less than 20 minutes to generate. Jobs were allocated 64GB of memory each.

Mistral SAE training was run on a single V100 GPU. Initially caching activations from Mistral 7B on one billion tokens took approximately 60 hours. Training the SAEs on the saved activations took another 36 hours.

H.3 ERROR BAR CALCULATION

In Fig. 6 we report 96% error bars for all intervention methods. To compute these error bars, we loop over all intervention methods and all layers and compute a confidence interval for each (method, layer) pair across all prompts. Assuming normally distributed errors, we compute error bars with the following standard formula:

$$EB = \mu \pm z * SE$$

where μ is the sample mean, z is the z score (slightly larger than 2 for 96% error bars), and SE is the standard error (the standard deviation divided by the square root of the number of samples). We use standard Python functions to compute this value.

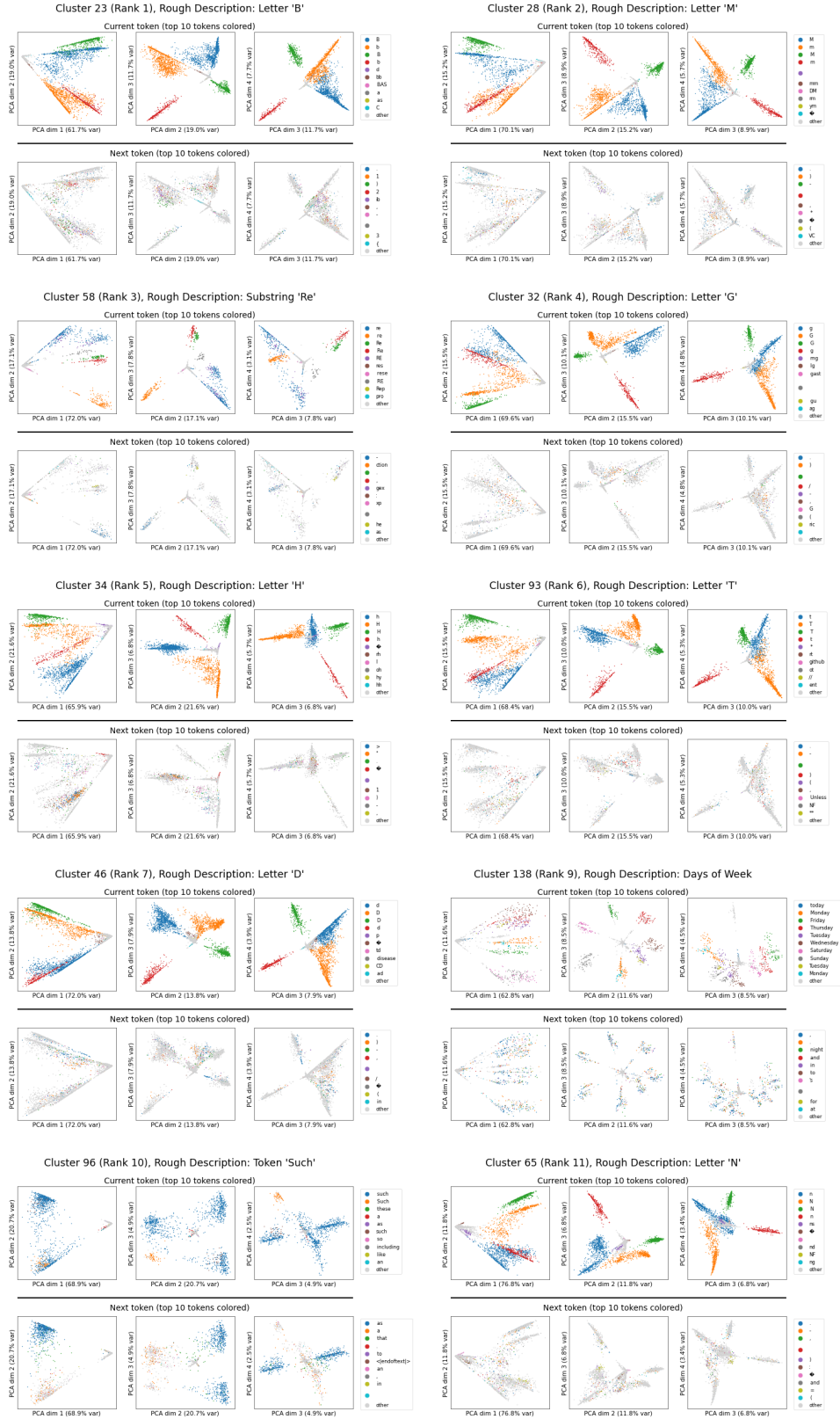


Figure 17: Top 10 GPT-2 clusters by Mixture and Separability Index.